

MAOTIN: A High-Accuracy and -Efficiency Network for Overloaded Truck Identification With GPS Trajectories

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Abstract—Overloaded truck transportation poses a significant threat to road safety. However, mainstream methods for overloaded truck identification suffer from low efficiency, limited monitoring range, and high costs, posing obstacles to curbing overloaded transportation. To tackle these challenges, this study proposes a novel approach called a multispect overloaded truck identification network (MAOTIN), which achieves accurate identification by analyzing truck trajectory data comprehensively. MAOTIN comprises three efficient and specific modules: an overloaded feature identification module (OFIM), a trajectory similarity identification module (TSIM), and a decision fusion module (DFM). Initially, the trajectory representation method was designed to transform the multidimensional spatiotemporal truck trajectory data into images, aggregating overload features while reducing dimensionality. Subsequently, the OFIM and TSIM are developed to comprehensively identify overloaded behavior from typical overloaded features and overloaded trajectories similarity features. Finally, the DFM adaptively and fully fuses feature images of different properties to obtain the ultimate identification result. To evaluate the performance of MAOTIN, this study conducted experiments on the constructed overloaded truck dataset. The results indicated that the model achieves an accuracy of 96.4% and 316 FPS, surpassing mainstream image classification networks. Moreover, MAOTIN performed a precision of 97.3% on practical application testing and assisted traffic police in apprehending 7591 overloaded trucks in one quarter, demonstrating its high practical value. Hence, MAOTIN enables accurate, rapid, and comprehensive identification of overloaded trucks and can be effectively applied to the overloaded identification tasks in urban traffic management.

Index Terms—Convolutional neural networks (CNNs), intelligent system application, overloaded truck identification, trajectory similarity.

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I. INTRODUCTION

ROAD transportation is a fundamental driver of economic and social development. However, the high profitability of overloaded transport makes it difficult to eliminate the issue of overloaded truck transportation [1]. Overloaded transport accelerates the degradation of critical infrastructure such as highways and bridges, increasing the risk of rollovers and loss of control, even severe road accidents [2].

Traditionally, overloaded trucks were identified through manual weighing at checkpoints. However, this method has proved inefficient and resource-intensive, and it failed to tackle the issue of trucks deliberately avoiding inspection by taking alternative routes. Recent studies have explored new weight-measuring technologies [3], which aim to automate the inspection process. It includes dynamic and vehicle-mounted weighing systems [4], [5]. The dynamic weighing method improves efficiency but suffers from limited inspection coverage. Vehicle-mounted sensors provide real-time monitoring [6], [7], but their high cost and complexity limit their practicality for large-scale deployment. On the other hand, traffic management systems now collect vast amounts of trajectory data from truck-mounted GPS devices. Despite its potential, this data remains underutilized in overloaded truck identification.

Vehicle trajectory data provides valuable insights into driving behavior, road conditions, and vehicle performance. Mining patterns from this data has proved feasible for tasks such as traffic mode identification [8], [9], [10], state prediction [11], [12], and traffic flow forecasting [13], [14]. However, applying to overloaded truck identification remains an area with limited exploration. The implicit characteristic of overload-related behaviors within trajectory data and the large volumes of data present significant challenges for current convolutional neural networks (CNNs), which struggle to capture these complex patterns.

To address these challenges, this article proposes a novel framework for overloaded truck identification based on trajectory analysis. This framework is specifically designed to capture the unique driving information of overloaded trucks. Compared with normal trucks, overloaded trucks exhibit distinct driving patterns to avoid inspection. Moreover, overloaded truck drivers often choose routes that maximize efficiency, leading to striking similarities in the trajectories [15]. Our framework integrates both typically overloaded

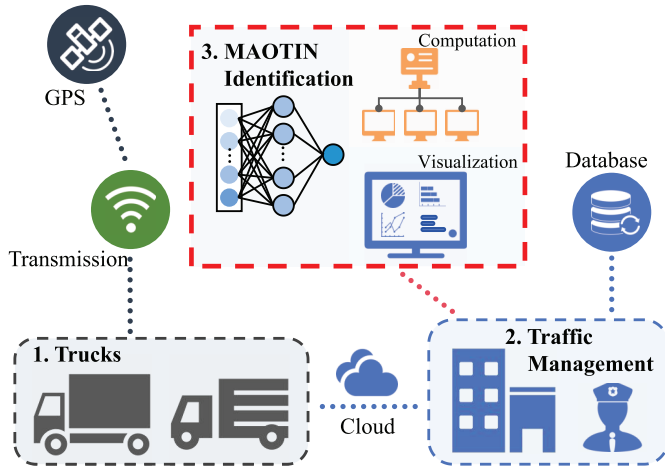


Fig. 1. Application process of MAOTIN. Initially, the GPS device affixed to the vehicle acquires real-time trajectory data. Subsequently, this data is relayed to the traffic management platform for recording and storage. Finally, MAOTIN identifies the overloaded trucks through the data, processes the information, and presents the visualization results in the system.

features and trajectory similarity features, resulting in the development of an accurate, efficient, and highly robust multispect overloaded truck identification network (MAOTIN). It consists of the overloaded feature identification module (OFIM), a trajectory similarity identification module (TSIM), and a decision fusion module (DFM). The OFIM captures typical overloaded features, the TSIM identifies trajectory similarity features, and the DFM merges the outputs from both modules to generate the final identification results.

The contributions of this article are as follows.

- 1) This article proposes a multispect overloaded truck high-precision identification network, a deep learning-based framework that comprehensively learns and extracts overloaded truck features from trajectories. To the best of our knowledge, MAOTIN is the first model to use driving trajectory data for overload identification.
- 2) This article introduces an innovative truck trajectory representation method that transforms multidimensional driving data into a 2-D image. This transformation enhances overload feature aggregation, simplifying the training process and improving network efficiency.
- 3) This article designs two specialized neural network modules for feature extraction. The OFIM fully extracts typical overloaded features from multiple spatial scales, while the TSIM analyzes the trajectory similarity features between current and historically severely overloaded trajectories. Together, these modules significantly improve the accuracy and robustness of the overload identification process.

A notable advantage of our model is its practical applicability. It can be seamlessly integrated into traffic management platforms to establish an intelligent overload detection system. Fig. 1 depicts the application process of MAOTIN. The GPS devices installed on trucks continuously transmit real-time trajectory data to the data cloud and traffic management platform database for recording and storage. Our model per-

forms comprehensive and rapid analysis of these trajectory data, generating accurate overload identification results. These results are subsequently visualized, providing crucial decision-making support for traffic police officers. This integrated approach enables timely detection and enforcement actions against overload violations.

The remainder of this article is organized as follows. In Section II, the article reviews the existing work related to the traditional overloaded truck identification and pattern recognition methods based on traffic trajectory data. Next, Section III outlines the methodology of our proposed MAOTIN. Subsequently, Section IV introduced the dataset construction and experimental settings, followed by the results analysis of the model testing experiment and practical application testing. Finally, the article concludes in Section V.

II. RELATED WORKS

A. Traditional Methods of Overloaded Truck Identification

The traditional method of identifying overloaded trucks relies on weighing sensors, which can accurately measure the weight to determine whether the truck is overloaded. Currently, most cities still use the manual weighing method, which requires traffic police to individually inspect trucks at fixed overload checkpoints (such as highway intersections or expressway checkpoints). The method suffers from a limited monitoring range, low detection efficiency, significant consumption of workforce and material resources, and inability to address the issue of trucks deliberately avoiding inspection by taking alternative routes.

Some researchers have improved the inefficiency of manual weighing and designed dynamic weighing devices. These devices automatically record the weight when trucks pass through the weighing devices at checkpoints at low speeds. The cameras capture images of the vehicle, recording details such as the number of axles, license plate number, and other essential information, enabling determination of whether the vehicle is overloaded. Qiao and Zhao [16] designed a dynamic weighing device for the road surface using the bending plate sensor with strain gauges. The principle of this method is straightforward, but its detection accuracy is compromised and demands a reduced passing speed. To overcome this problem, Lydon et al. [17] improved the system by adding fiber-optic sensors, a vision system, and a new shear strain transducer, enhancing weighing system accuracy but raising device cost. While the above dynamic weighing method improves detection efficiency and reduces workforce costs, it is limited to fixed points, resulting in a small. Consequently, it failed to solve the problem of trucks evading inspection by making detours.

Some researchers have sought to overcome the limitation of fixed monitoring points by developing vehicle-mounted load cells for real-time weight monitoring. Vehicle-mounted sensors operate on different principles. For example, Gaira et al. [18] installed laser displacement sensors on trucks to measure the deformation of the steel plate spring, which is then used to estimate the truck's weight. While the approach is straightforward, it faces challenges in accurately estimating

weight, as extensive calibration is needed to correlate spring deformation with weight, and the results are sensitive to external environmental factors. Similarly, Wang et al. [19] used onboard sensors to measure road gradient and rear suspension deformation for weight estimation. Although this method can be applied to most trucks, it requires precise sensor installation and is costly. Despite enabling real-time monitoring of truck overload behavior, vehicle-mounted weighing systems are expensive and face limitations in terms of truck compatibility. No universal vehicle-mounted weight monitoring device has been developed, limiting the scalability of these methods.

B. Pattern Recognition Method Based on Traffic Spatiotemporal Trajectory Data

Spatiotemporal trajectory data captures a wealth of behavioral features, and deep analysis of this data provides strong support for decision-making and solving complex tasks. With advancements in high-precision positioning technologies and the Internet of Things (IoT), large volumes of valuable traffic trajectory data have become readily available, enhancing the accuracy and effectiveness of traffic identification and prediction tasks.

Analyzing and mining the information from trajectory enables the acquisition of the behavioral characteristics of the vehicle ontology. This process facilitates trajectory classification, anomaly detection, and prediction. Ma et al. [20] proposed a multiattribute-based fusion network (MASO-MSF) and constructed an MASO structure to represent the local motion state and spatial characteristics of a GPS trajectory segment at different spatial scales. Additionally, they designed a multistage fusion model to accurately identify traffic modes, which fully considers the spatial scale information of the features. However, the drawback of the model is complexity, which results in poor performance when processing large volumes of trajectory data. Yu and Huang [21] proposed a deep learning framework for driving trajectory anomaly detection, STDTB-AD. It divides the road network into a series of spatiotemporal units and uses the motion parameters of each spatiotemporal unit to represent the driving mode and characteristics. Then, accordingly, to sense abnormal driving behavior. Ji et al. [22] introduced a spatiotemporal self-supervised learning framework for traffic flow prediction, which enhances the model's ability to represent spatial and temporal heterogeneity. This is achieved through adaptive data augmentation and supported by self-supervised learning tasks. The above methods accomplish pattern recognition by designing various network structures to represent trajectory features. However, it is challenging for these networks to handle overloaded truck identification. Unlike other traffic data, truck journeys often span long distances, resulting in large volumes of data that cannot be directly processed as simple time series. Additionally, the overloaded characteristics are implicitly involved in the driving data, which cannot be simply described by the shape of the trajectory. The features behave differently at different spatial scales, which also complicates the learning process for conventional CNNs.

Unlike the previous methods, an accurate, efficient, and highly robust multispect overloaded truck high-precision

identification network is proposed in this article. The trajectory representation method fully considers the truck driving characteristics and converts the driving trajectory data with the image. The OFIM and TSIM form two branches of the network. They identify overload from typical overloaded features and trajectory similarity features, respectively. The OFIM acquires multiscale and accurate feature maps, and the TSIM efficiently gets the trajectory similarity feature map. Both of them are merged in the DFM for the last results. This design introduces the specific characteristics of overloaded trucks into the model's structure, enhancing the accuracy and reliability of the results. In addition, the trajectory representation method helps MAOTIN apply to massive truck data processing tasks.

III. METHODOLOGY

A. Overview of MAOTIN

Fig. 2 shows the overall structure of MAOTIN. The process begins with the trajectory representation method, which transforms truck driving trajectory data into an image, serving as the model's input. Then, the OFIM and TSIM identify overload behavior from different aspects. The OFIM mines multiscale typical overloaded features layer by layer. Its unique Swin feature pyramid (SwinFP) stage and feature selection layer enable it to capture richer and more comprehensive feature maps efficiently. Meanwhile, the TSIM analyzes the similar features of driving routes and historically overloaded routes. It generates a similarity feature map using a full convolutional Siamese network (FCSN). Following this, the DFM comprehensively integrates the two feature maps, yielding accurate overloaded identification results.

B. Trajectory Representation Method

The initial truck driving trajectory data is multidimensional spatiotemporal data (including time, position, velocity, etc.), presenting challenges due to its large volume, dimensionality, and feature dispersion. For the above problems, a trajectory representation method is proposed. It can convert complex truck driving trajectories into 2-D trajectory images.

The trajectory representation method first defines the range and resolution of the trajectory image, mapping the input truck trajectory data into the spatial domain. A transformation model is then established to convert trajectory information into image pixels, where the number of trajectory points is encoded as pixel gray. After this, the method reinforces the essential features in the generated image, and the final truck driving trajectory image is produced. The overall process is depicted in Fig. 3.

Fully analyze the truck's driving characteristics. An excessive number of points in a pixel indicates that the truck is stationary, with limited observable feature information. Fewer points, particularly when distinct from surrounding pixels, suggest dynamic behaviors such as deceleration, turning, lane changing, or potential overloads, making this feature the most informative for overload identification. Sparse trajectory points indicate high-speed travel, with trajectory shapes providing

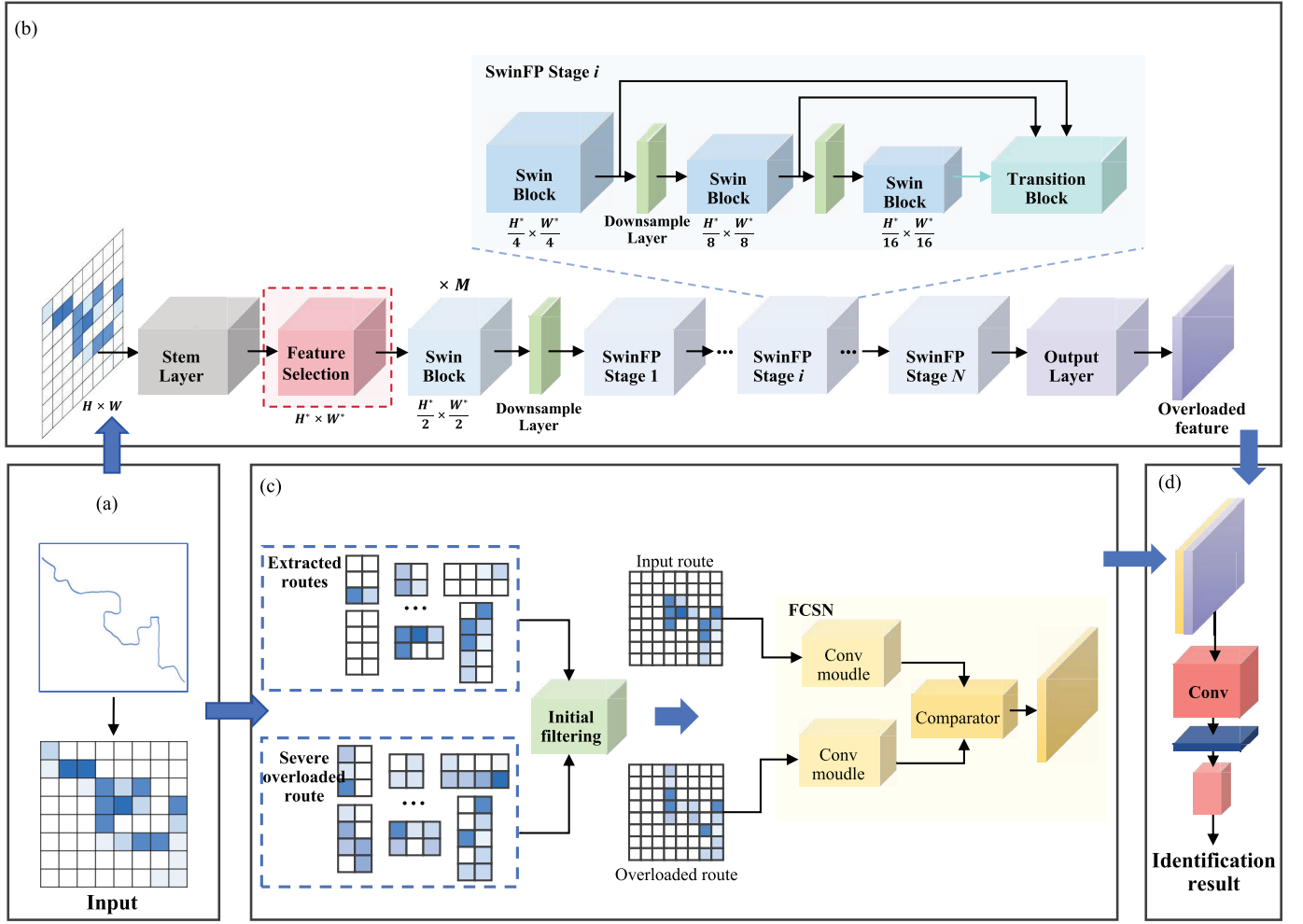


Fig. 2. Overall structure of MAOTIN. The figure comprises four distinct parts. (a) Tasked with converting trajectory data into trajectory images to serve as the MAOTIN's input. The remaining three parts represent the main components of MAOTIN. (b) OFIM and (c) TSIM constitute two branches of the MAOTIN, each executing overloaded identification from unique aspects. Lastly, (d) DFM extensively integrates the identification outcomes from both branches.

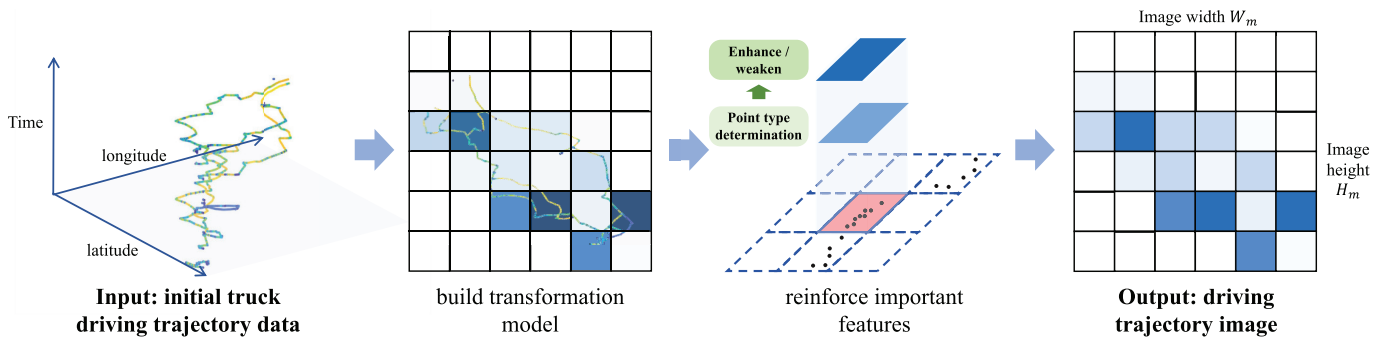


Fig. 3. Truck driving trajectory image generation process. A driving trajectory image is utilized to represent the initial driving trajectory data.

more significant features. The study summarizes this into two types of pixels: differential pixels, which show significant differences from neighboring regions and are enhanced to highlight key features and stay pixels, representing areas with excessive points that should be suppressed in the image to prevent the network from overly concentrating on these regions. Considering these, the method offers special processes from global and local perspectives.

The first is in the stage of building the transformation model, converting the trajectory information into

$$g_{i,j} = \log(1 + \alpha * N_{i,j}) \quad (1)$$

$$g_{i,j}^* = 255 * \text{Norm}(g_{i,j}) \quad (2)$$

where i and j denote the pixel coordinates, $N_{i,j}$ is the number of trajectory points at that pixel, $g_{i,j}^*$ represents the gray value, and $g_{i,j}$ is the unnormalized value. The hyperparameter α adjusts

the sensitivity of the gray value to the number of trajectory points. The model differs from the normal linear transformation [23], [24]. Its unique functional properties reduce the impact of stay pixels. The special mapping relationship introduced by the prior information is leveraged to enhance texture features globally.

The local feature enhancement is involved in the feature reinforcement stage. The individual pixels are classified into differential pixels and stay pixels based on their density and predefined thresholds. The type of each pixel determines whether it should be enhanced or suppressed. The enhancement and suppression factors are defined as

$$E_{en} = k \cdot \exp(\tau \cdot (N_i - N_{neighbor})) \quad (3)$$

$$E_{we} = \left(1 - \frac{1}{1 + \exp(\lambda(N_i - T))}\right) \quad (4)$$

where k is a constant factor, γ controls enhancement intensity, λ governs the width of the transition, and T is the threshold for identifying stay pixels. These factors are applied to the respective pixels, resulting in an enhanced trajectory image that emphasizes the truck's driving characteristics.

C. Overloaded Feature Identification Module

Typical overloaded features exhibit significant disparities across various spatial scales within truck trajectory images. Its complexity also brings obstacles to the learning process of the conventional CNN. In this regard, this study designed the OFIM to quickly and thoroughly mine the typical overloaded features in truck driving trajectory images.

Fig. 2(b) illustrates the architecture of the OFIM. The input, an overloaded vehicle trajectory image, first undergoes feature extraction through a stem layer and N consecutive Swin blocks. This stage generates high-resolution feature maps. These feature maps are then passed through M -cascaded SwinFP stages, which progressively expand the receptive field, enabling the network to capture features at multiple spatial scales. As a result, the network produces a comprehensive feature map highlighting the overloaded characteristics across details of various scales.

It is important to note that vehicle trajectory images differ from traditional images. The valuable information within these images occupies a smaller portion of the frame, and background texture information is weak, providing limited utility for overload identification. Thus, the model incorporates a feature selection layer in the backbone network to enhance efficiency and identification performance.

The feature selection layer divides the feature map into blocks and evaluates the importance of the information in each block. Based on these evaluations, a mask is generated and applied to the original feature map, filtering out unimportant features. Considering computational resources, the model assesses information strength within each block using variance. This feature selection process reduces redundant information and improves identification speed. The variance is calculated as

$$\sigma_{c,m,n}^2 = \frac{1}{P^2} \sum_{i=1}^P \sum_{j=1}^P (X_{c,m,n}^{(i,j)} - \mu_{c,m,n})^2 \quad (5)$$

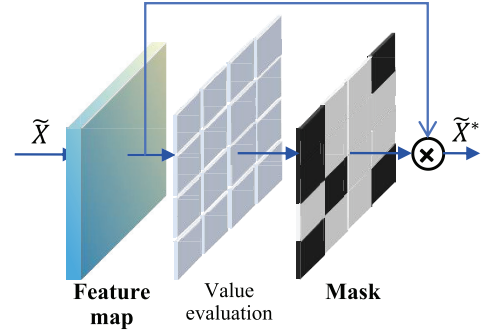


Fig. 4. Feature selection mechanism.

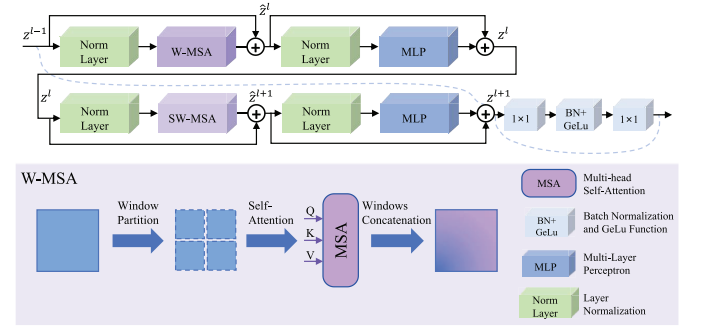


Fig. 5. Structure of the Swin block.

where m and n represent the block's position, c is the channel index, and μ is the mean value of the block. If the variance falls below a threshold τ , the block is deemed information-weak and selected for filtering. The corresponding mask positions are marked accordingly. As shown in Fig. 4, the resulting mask is applied to the input feature map to generate a filtered feature map.

The stem layer of the OFIM consists of a basic 3×3 convolution module. The core feature extraction unit of the OFIM is the SwinFP stage, which employs an encoder-decoder structure. The encoder Swin blocks extract multiscale features, while the decoder transition block synthesizes them into a single-scale, high-resolution output. High-level (low-resolution) features from earlier stages guide the learning of low-level features in subsequent stages, improving the model's ability to capture both fine-grained and global features.

The structure of the Swin block is shown in Fig. 5. It is referred to as the Swin Transformer block when designing [25]. Two skip connections are incorporated to fuse low-level and high-level features, enhancing the model's ability to capture richer semantic information. The Swin block processes image data via a self-attention mechanism applied within local windows, progressively capturing local and global information. Compared to standard convolutional blocks, this approach significantly enhances the network's capacity to handle complex image features.

With the aid of the OFIM, the typical overloaded feature information of multiple spatial scales is efficiently extracted from the truck trajectory image. The information is then represented as a feature map, serving as the primary basis for high-precision overloaded identification.

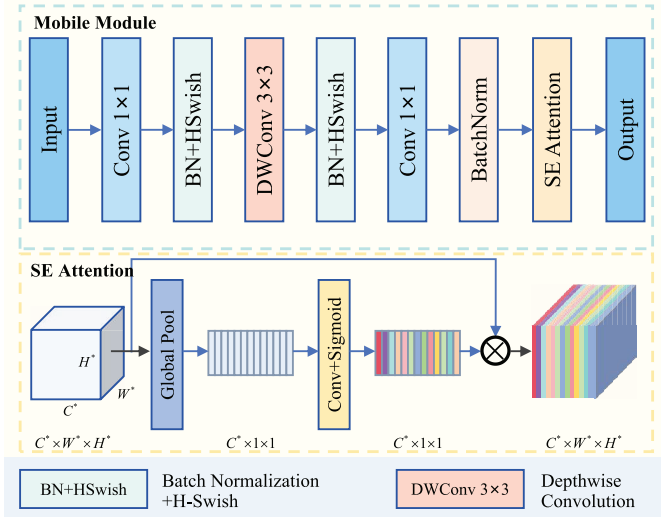


Fig. 6. Detail of the mobile and SE attention modules.

D. Trajectory Similarity Identification Module

The TSIM extracts the trajectory similarity feature maps between the historically severely overloaded road sections and the corresponding driving routes in the truck trajectory images. Based on this consideration, this study designed the TSIM to extract the similarity information efficiently.

As shown in Fig. 2(c), the extracted routes are paired with historical routes, forming route pairs. Pairs with low similarity are initially filtered out, indicating that the truck did not pass through those areas. The remaining pairs are used to generate feature maps for similarity calculation. The core of the TSIM is an FCSN. It consists of two identical branches with the same network structure and shared weights. These trajectory pairs are processed through the two branches to extract feature maps, and a comparator is employed to assess the similarity between the two feature maps.

For feature extraction, FCSN primarily utilizes the lightweight mobile modules [26]. Building upon this, the model incorporates the squeeze-and-excitation (SE) attention modules [27] to capture channel-wise dependencies and enhance model performance. The details of its design are shown in Fig. 6. The cosine similarity is designed as the comparator, providing the similarity information on the feature maps. Given two input feature maps, $\mathbf{F}_1$ and $\mathbf{F}_2$, the output similarity feature map $\mathbf{F}_{\text{sim}}$ is calculated as

$$\mathbf{F}_{\text{sim}_{ij}} = \frac{\mathbf{F}_{1_{ij}} \cdot \mathbf{F}_{2_{ij}}}{\|\mathbf{F}_{1_{ij}}\| \|\mathbf{F}_{2_{ij}}\|}. \quad (6)$$

In this equation, (i, j) represents the spatial coordinate in $\mathbf{F}_{\text{sim}}$. The feature map generated by the TSIM and the feature map generated by the OFIM are then transferred to the DFM for final overload behavior recognition.

E. Decision Fusion Module

The DFM module is designed to effectively combine feature maps extracted from the OFIM and TSIM. The structure of the DFM is shown in Fig. 7. The DFM module consists of two

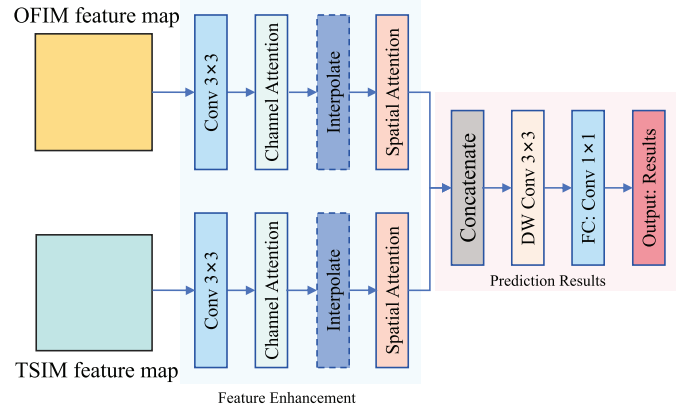


Fig. 7. Structure of the DFM.

TABLE I
DETAIL INFORMATION OF THE DATABASE

Attribute	Value
Dataset size	240 GB
Number of trucks	492,001
Number of information	2.5 billion
Number of overloaded trucks	150,225
Number of normal trucks	341,776
Duration of travel time	November 1, 2023 -November 30, 2023

branches, each processing a different type of feature map to ensure that the output feature maps have the same dimensions for concatenation. After fusion, the features are passed through a simple classifier to produce the final prediction. The module incorporates both spatial and channel attention mechanisms to selectively enhance the most relevant features. Additionally, the DFM module employs depthwise separable convolution in the classifier, significantly reducing computational complexity while preserving the model's representational power.

IV. EXPERIMENT AND TESTING

To validate the multiaspect overloaded truck high-precision identification network, a large-scale truck overload dataset, and an experimental platform were constructed. The experiments were conducted on the dataset. In addition, actual application testing was performed to demonstrate the practicability of the model.

A. Experimental Preparation

1) *Dataset Construction*: The database utilized in our study comes from the Tianjin Public Transportation Big Data Platform, China. The data is collected by GPS devices installed on the trucks, which regularly transmit information to the platform for storage. The data attributes include license plate number, driving time, longitude, latitude, and speed. The detailed information of the database is in Table I. All the behaviors of trucks in the database have undergone confirmation of their overload status through the manual checkpoint weighing method. The time range is 24 h before the inspection.

A portion of the original file of truck driving trajectory data is depicted in the blue section (Fig. 8). Its data was optimized and preprocessed based on the computing cluster.

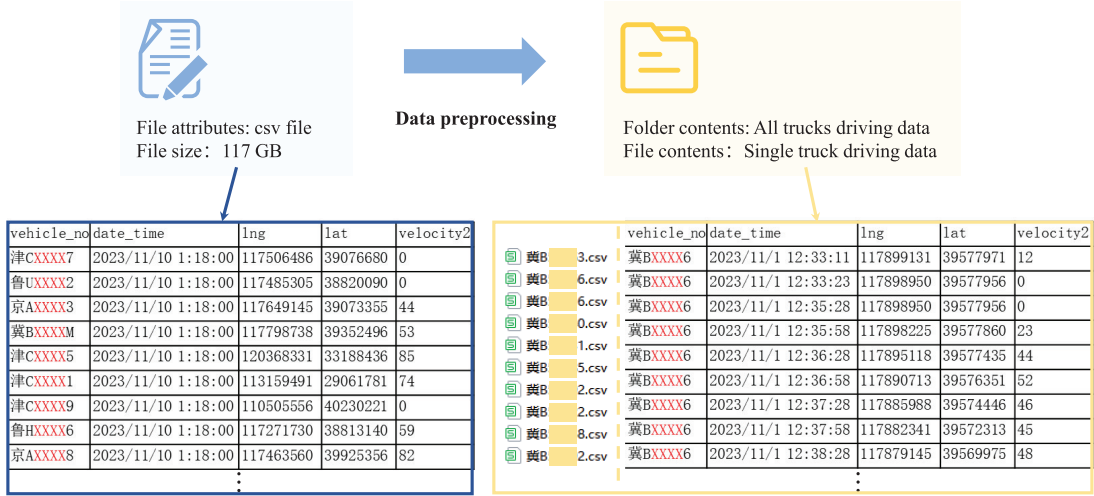


Fig. 8. Original and optimized data file.

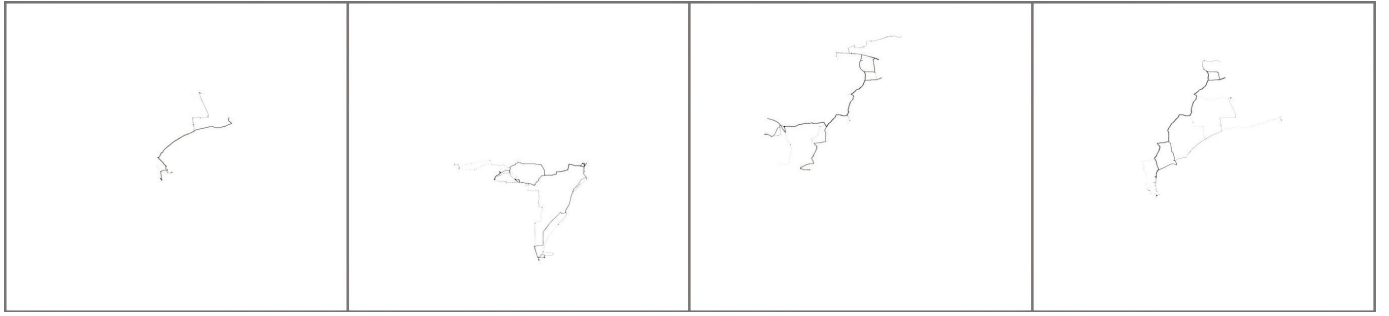


Fig. 9. Transformed truck driving trajectory image.

The yellow portion in Fig. 8 illustrates the optimized data file. As shown in Fig. 9, the processed data files are converted into truck trajectory images according to the method proposed in Section III-B. For ease of differentiation, Fig. 9 displays the figure after color inversion, where the darker areas in the figure correspond to higher gray values. The trajectory of the truck is clearly visible in the trajectory image, with important trajectory segments enhanced. The pixel position of the trajectory in the image represents the actual location of the trucks, with each pixel corresponding to an approximate spatial range of 200 m. This approach not only preserves key trajectory information but also allows for dimensionality reduction of the trajectory data. The images are classified into overloaded and normal categories. It includes 200000 truck trajectory images, with a ratio of overloaded trucks to normal trucks is 1:1. 90% of the images in the dataset form the trainset for model training, while the rest form the test set to verify the model's validity.

It can be concluded from the trajectory data that, compared with normal trucks, overloaded trucks exhibit driving characteristics such as detours, frequent lane changes, driving on paths to avoid inspection at overloading control points, and engaging in illegal transportation. Fig. 10 displays the abnormal driving trajectory of some overloaded trucks. Fig. 11 visualizes the driving trajectory heatmap of some overloaded trucks. The horizontal and vertical coordinates represent the

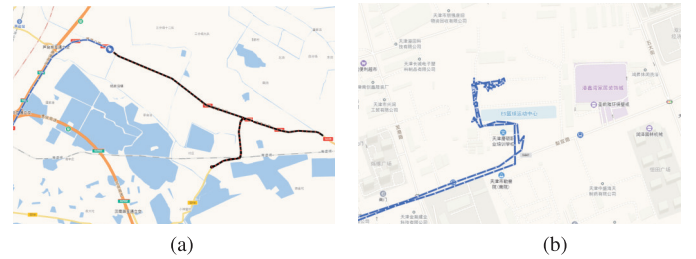


Fig. 10. Abnormal driving trajectory of some overloaded trucks. (a) Making detours. (b) Changing roads more often and driving on paths.

position of pixel points in the truck driving trajectory image. The color of each pixel denotes the driving area heat (trajectory point density) at that location, with the color bar ranging from dark to light, indicating increasing heat from low to high. Some high-frequency truck driving routes can be clearly seen in Fig. 11, which shows that the overloaded trucks prefer specific routes, leading to high similarity trajectories across different trucks. Preliminary trajectory data analysis proves the feasibility of overloaded identification by mining typical overloaded characteristics and trajectory similarity.

2) *Experimental Platform Establishment*: All experiments are conducted in a workstation with an Intel Core i7-11700 CPU @2.50 GHz, 32 GB memory, and two NVIDIA RTX

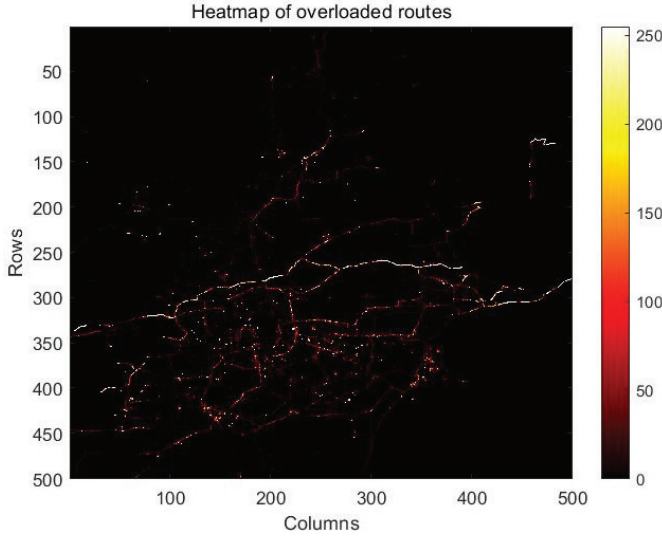


Fig. 11. Driving trajectory heatmap of some overloaded trucks.

4090 GPUs. The operating system is Ubuntu 20.04. Deep learning models are implemented in the Python programming language.

In addition, a big data computing cluster was built to preprocess massive truck trajectory data, consisting of three host computers with AMD Ryzen 9 5950X 16-Core Processor CPU, 128 GB RAM, and CentOS 7 64-bit system.

3) *Historically Severely Overloaded Database Establishment*: The work summarized the driving routes of severely overloaded trucks and combined them with the investigation and enforcement of traffic management departments to establish a historically severely overloaded truck database. It contains 5807 severely overloaded trucks (exceeding the load capacity by more than 100%) and forms 20 severe cargo loading areas and 71 seriously overloaded road sections. Part of the information in the historically severely overloaded database is shown in Table II. The database provides data support for the TSIM in Section III-D.

B. Model Testing Experiment

1) *Model Training Details*: The model was trained using the constructed training set. First, the upper and lower branches were trained separately. Then, the weights of both branches were fixed, and the full model was assembled. Finally, the complete model underwent training. The model is trained using the stochastic gradient descent (SGD) optimizer with an initial learning rate of 0.0001, a momentum of 0.9, and a weight decay of 0.0004. The learning rate is dynamically adjusted following a warm-up and exponential decay strategy: it linearly increases from 1×10^{-5} to 4×10^{-4} over the first ten epochs, remains constant for the next five epochs, and subsequently decays exponentially with a factor of 0.8 for the remaining training period. The model is trained for 50 epochs with a batch size of 32. Input images are resized to 500×500 pixels. Data preprocessing includes normalization with a mean of 0.485 and a standard deviation of 0.229. The loss function is the cross-entropy criterion. The training

TABLE II
PART OF THE INFORMATION IN THE HISTORICALLY SEVERELY OVERLOADED DATABASE

Overloaded truck license plate number	Cargo loading areas	Overloaded driving routes
Tianjin CXXXXX8 Tianjin CXXXXX6 Hebei BXXXXX7	Shangcang Town, Tianjin, China	Pingxiang Line - Tangtong Line - Jinwei Highway
Tianjin CXXXXX3 Hebei BXXXXXS Hebei JXXXXX6	Yinliu Town, Tianjin, China	Jingfu Line - Jinghao Road - Jingha Highway
Hebei EXXXXX5 Hebei BXXXXXV Hebei BXXXXX2	Jingbin Industrial Park, Tianjin, China	Jinglan Line - Donghuan Road - Langwu Road - Guda Road
Hebei JXXXXX3 Hebei BXXXXX5 Tianjin CXXXXX2	Zhongtang Town, Tianjin, China	Sanshen Line - Jin Shen Line - Ge Wan Line - Zhonggang Road
Hebei BXXXXXW Hebei BXXXXXX Hebei JXXXXX5	Jinjiatai Village, Tianjin, China	Drainage line - Bao Wu line - Daxin Road - Lin Da line
Tianjin CXXXXX5 Hebei EXXXXX9 Hebei BXXXXXW	Daqiuzhuang Industrial Zone, Tianjin, China	Jingba Line - 310 Provincial Highway - 110 Provincial Highway - Tuanwang Line - Gangjing Line
Tianjin EXXXXX5 Tianjin CXXXXX9 Tianjin CXXXXX2	Sijiatuo, Tianjin, China	Jiuyuan Line - Provincial Highway 205 - Bao Lu Highway - Hannan Road

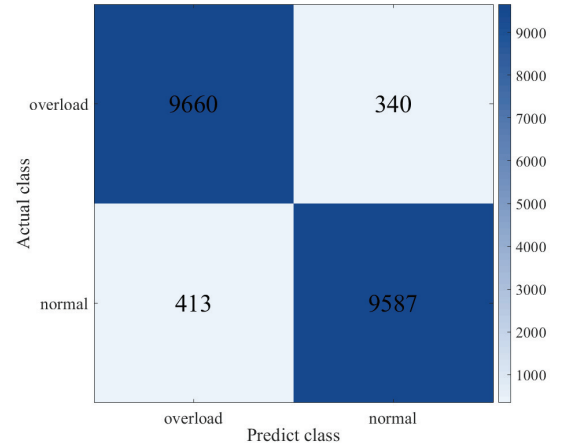


Fig. 12. Confusion matrix of test results.

process applied a ninefold cross-validation method. Model performance is evaluated using an independent validation set, and the best-performing model is saved based on validation accuracy.

2) *Model Testing Results*: The model was tested on the test set. The results show that the accuracy of MAOTIN is 96.4%, the recall is 96.6%, and the precision is 95.9%. The confusion matrix is shown in Fig. 12.

The comparison experiments were conducted with mainstream image classification networks, and the results are presented in Table III. All compared networks were trained with the same training dataset as our proposed network, employing the trajectory image generation method introduced in Section III-B. This study's model performance evaluation

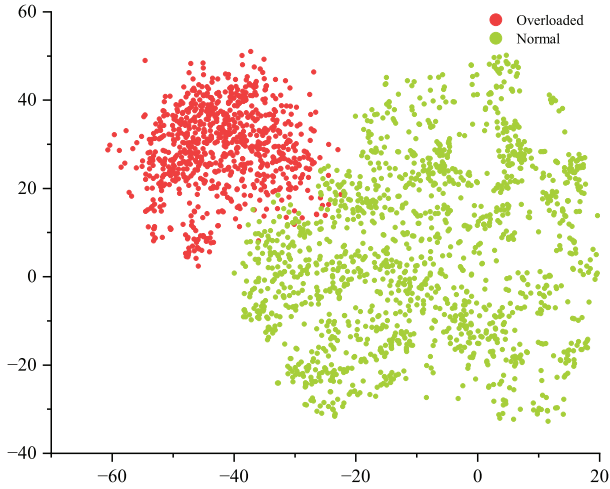


Fig. 13. t-SNE visualization of feature maps.

TABLE III
COMPARISON EXPERIMENTS RESULTS

Model	Accuracy	FPS	GFLOPs	GMacs	Parms (M)
MAOTIN(Proposed)	96.4%	316	19.58	9.79	22.12
SVM [28]	69.2%	41	23.12	11.59	8.65
VGG16 [29]	80.7%	138	152.39	76.11	138.36
Resnet-101 [30]	79.8%	568	81.75	40.75	44.55
Densenet-121 [31]	83.4%	104	28.16	13.96	6.96
Inception-v3 [32]	82.8%	146	39.51	18.71	27.26
Squeezenet-1_1 [33]	84.5%	340	6.84	1.90	1.24
Shufflenet-v2_x1.5 [33]	85.9%	250	3.09	1.53	3.50
Mobilenet-v3 [26]	82.8%	319	0.58	0.28	2.54
Mobilenet-v3_large [26]	85.6%	270	2.26	1.10	5.48
Ghostnet [34]	84.6%	313	1.49	0.72	5.18
Regnet-x_16gf [35]	88.1%	222	16.64	8.26	9.19
Efficientnet-b7 [36]	86.2%	206	4.07	1.98	5.29
Repvgg [37]	88.4%	275	15.69	7.82	7.83
Convnext [38]	89.4%	161	150.2	74.97	88.59
Convnext-tiny [38]	88.5%	289	43.46	21.67	28.59
Swin Transformer [25]	89.2%	257	15.4	7.66	28.29
FlashInternImage-T [39]	89.7%	281	4.46	2.23	30.00
FasterViT-1 [40]	89.5%	217	5.3	2.71	53.40
ViTamin-L [41]	89.4%	194	13.2	6.6	123.7

indices are accuracy, FPS, GFLOPs, GMacs, and Parms. The optimal results are marked in bold red.

In the experiments, MAOTIN achieves the highest accuracy compared to all other mainstream methods. FPS and FLOP results of MAOTIN show that it can perform identification with high efficiency and accuracy when dealing with massive truck driving data. The traditional machine learning method SVM performs the worst in the experiments, demonstrating that it is unsuitable for overloaded truck identification. Earlier CNN models VGG16, ResNet-101, Densenet-121, and Inception-v3 have improved compared to SVM but are far from the requirements. MAOTIN overmatches these models in identification accuracy, speed, and computational power consumption. The lightweight networks, Squeezenet, Shufflenet, Mobilenet, and Ghostnet, have better identification speed and computational power consumption

TABLE IV
ABLATION EXPERIMENT TEST RESULTS

Model	OFIM	TSIM	DFM	Accuracy
OFOTIN	✓	×	×	90.7%
TSOTIN	×	✓	×	78.6%
MBOTIN	✓	✓	×	95.1%
MAOTIN	✓	✓	✓	96.4%

The "×" indicates that the models displayed in the first column do not contain the corresponding module; the "✓" represents the presence of the modules.

TABLE V
KEY COMPONENT COMPARISON AND ABLATION EXPERIMENTS

Model	Accuracy
MAOTIN	96.4%
trajectory representation method in [24]	91.7%
trajectory representation method in [23]	92.3%
OFIM without feature selection layer	94.5%

performance, but the accuracy is much lower than MAOTIN. In comparison, the accuracy of MAOTIN increased by 11.9%, 10.5%, 10.8%, and 11.8%, respectively. Compared with the latest CNNs such as Regnet, Efficientnet, Repvgg, and Convnext, MAOTIN has a clear advantage in terms of accuracy and outperforms these models in identification speed and computational power consumption. Compared to the most advanced Transformer-based models—Swin-T, FlashInternImage-T, FasterViT, and ViTamin—MAOTIN still maintains a significant advantage in terms of accuracy. Among the reference models, FlashInternImage-T achieved the highest accuracy results. However, MAOTIN exhibits superiority over FlashInternImage-T, with an accuracy improvement of +6.7%, an increase in FPS by +35, and a reduction in Parms by −7.88. The results show that MAOTIN has the highest accuracy (96.4%) and the fourth fastest speed (316 FPS), which poses an excellent compromise between identification accuracy, speed, and computational consumption.

3) *Ablation Experiment*: To better analyze the effectiveness of each module in MAOTIN, three diverse models were designed and tested, and the test data are displayed in Table IV.

The OFOTIN, TSOTIN, and MBOTIN are the models that replace the corresponding modules of MAOTIN. Compared to MAOTIN, the identification accuracy of the model OFOTIN (removal of the TSIM), the model TSOTIN (removal of the OFIM) decreased by 5.7% and 17.8%, and the model MBOTIN (replace the DFM with simplest full connected layer) decreased by 1.3%, respectively. This result indicates that the upper and lower branches of MAOTIN can mine overloaded features from different aspects and play a role in accomplishing the overloaded identification task. In contrast, the DFM module is used to fuse and classify different feature maps rather than to extract features. Moreover, the OFIM plays the leading role, and the TSIM plays the supporting role. There are severe limitations when relying on the trajectory similarity (TSIM) independently.

In addition, the tests of the trajectory representation method and feature selection layer also have been done. The results are shown in Table V. It shows that the proposed trajectory

TABLE VI
PRACTICAL APPLICATION TEST RESULTS

Method	Number of tested trucks (a quarter)	Predicted overloaded trucks	Actual overloaded vehicle	Precision
Proposed method	110457	7802	7591	97.3%
Manual weighting method	40979	-	4484	-

representation method is more applicable to the identification task of truck overload, and the accuracy is improved by 4.7% and 4.1%. After adding the feature selection layer, the model's accuracy is enhanced by 1.9%, indicating that it helps the model efficiently obtain the overload feature information from the feature map.

4) *Feature Visualization Experiment*: To evaluate the feature extraction capabilities of our model, we employed t-SNE [42] to visualize the high-dimensional feature space extracted by MAOTIN. The features were extracted before entering the classifier and reduced to two dimensions using t-SNE. The perplexity was set to 30, the learning rate to 200, and the number of iterations to 1000.

As displayed in Fig. 13, the resulting visualization shows a clear separation between classes, with minimal overlap, indicating that the model has learned highly discriminative features. The distinct clustering of red and green points further demonstrates the model's ability to effectively differentiate between categories. These results suggest that the network captures robust feature representations even before the final classification layer, reinforcing the effectiveness of the learned feature space for accurate identification.

C. Practical Application Testing

MAOTIN was integrated with the traffic management big data platform and deployed in traffic enforcement operations in Jizhou District, Tianjin, China. The model identified the overloaded trucks in a wide range to complete the initial screening work and provided precise enforcement areas for traffic policies to stop overloaded transportation violations effectively. The results of the actual application testing from January to March 2024 are shown in Table VI. It is important to note that our model undergoes daily updates based on the most recent overloading conditions and data from traffic management, ensuring its real-time accuracy and reliability.

This quarter, MAOTIN identified 110457 trucks and helped traffic police apprehend 7591 trucks. The actual identification precision is 97.3%, proving it can be adequate for overloaded identification in the real world. Compared with the traditional manual checkpoint weighing method, MAOTIN identifies 2.7 times the number of trucks. More importantly, it helps the police catch a large number of overloaded trucks that intentionally avoid inspection. Our method boasts rapid speed, high accuracy, and extensive coverage. On the other hand, as our method does not require the installation of additional sensors, it holds significant application value.

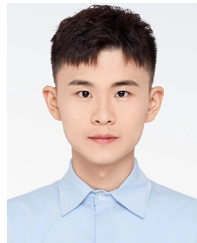
V. CONCLUSION

This study introduces MAOTIN, a deep-learning network designed for overloaded identification using trajectory data. The proposed multiaspect identification framework, customized trajectory representation method, and three efficient modules significantly ensure the accuracy and efficiency of MAOTIN. The innovative trajectory representation method converts driving trajectory data into images, enabling efficient feature aggregation while preserving critical overload patterns. The OFIM and TSIM identify overloads from different aspects. The OFIM leverages a unique SwinFP stage and feature selection layer to extract multiscale, typical overloaded features, while the TSIM utilizes a fully convolutional Siamese network to extract trajectory similarity features with the historically overloaded routes. Finally, the DFM integrates the feature maps from both the OFIM and TSIM, producing highly accurate overload identification results. Experimental evaluations on a constructed overloaded truck dataset show that MAOTIN achieves 96.4% accuracy, surpassing conventional image classification networks. Moreover, real-world deployment testing demonstrated a precision of 97.3%, illustrating its practical value for addressing truck overloading issues. Future work will focus on enhancing the model's performance by incorporating additional data sources, such as urban road data, points of interest (POI) data, and sensor data, to further refine overload identification. These advancements will enable MAOTIN to play a critical role in intelligent transportation systems.

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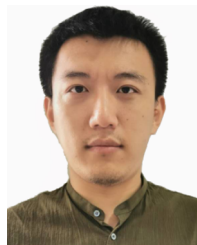
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